# Nathan Zhang

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## Education

**The University of British Columbia** Sep 2023 – May 2028
B.A., Combined Major in Economics and Statistics, Minor in Data Science **Cumulative average: 93.5/100**
**Selected coursework:** Statistical Inference, Machine Learning and Data Mining, Advanced Econometrics, Time Series and Forecasting

## Experience

**ESG Research Analyst Co-op** Sep 2026 – Present
British Columbia Investment Management Corporation (BCI) | Capital Markets & Credit Investments

- Investigate how executive compensation aligns management incentives with shareholder interests, helping the investment team prioritize governance risk factors. Independently construct variables from Capital IQ and ISS ECA and build Databricks pipelines for fundamental investment research.
- Automate voting rules, reconstruct ballot-time information, and reproduce research findings. Backtest against historical votes and use correlation analysis of subsequent risk outcomes to help the proxy team assess voting strategy.

**Research Assistant** Apr 2026 – Present
UBC Sauder School of Business | Supervisors: J. Frank Li and Cheng Chen

- As sole RA, build technological-exposure measures from CPS, O*NET, and USPTO data. Validate PatentsBERTa measures using Qwen comparisons and manual reading of task/patent samples.
- Fit weighted FE OLS and GMM models of wages and occupational sorting, selecting fixed effects. Draft the empirical section and maintain R/Python replication and Bash/Slurm HPC workflows.

**Research Assistant** May 2026 – Aug 2026
UBC Sauder School of Business | Supervisor: Allen Hu

- Assessed financial-news coverage by matching a ProQuest corpus of 3.2M+ articles against RavenPack at the article level and examining missingness.
- Independently built yt-dlp workflows for multimodal financial information extraction and analysis: metadata-to-panel generation, video/audio archiving and deduplication, and subtitle-availability checks.

**Research Assistant** Jan 2025 – May 2026
Vancouver School of Economics at UBC | Supervisors: Jonathan Graves and Laura Nelson

- Led CV/NLP/NLI teaching proposals. Co-developed a Kouroi notebook testing whether image embeddings recover art historical categories, implementing ConvNeXt V2 training/classification and writing the method explanations.
- Co-developed a tutorial comparing zero-shot stance predictions with human labels, in first revision at Programming Historian. Designed and taught a computer vision workshop.

## Selected Research

**Sequential Tone Residualization and Abnormal Returns in Earnings Calls** 2026
Joint first author. Accepted at ICAIF 2026, selected for oral presentation. Proposed the earnings-call application to test whether unexpected management tone predicts returns. Built a Capital IQ/IBES/CRSP event panel and co-designed information sets reflecting financial controls and prior dialogue at each call stage. Proposed dictionary/LLM validation of tone measures. Out-of-sample tests linked answer surprise positively to subsequent cumulative abnormal returns, with pre-event robustness checks.

**Memory Response Paths and Path-Structured Context Compilation** 2026
Joint first author. Submitted to AAAI, under review. Proposed response paths to trace a memory's effect on LLM answers over time by comparing matched continuations with and without it. Built benchmarks and model API code and analyzed results across models and settings. In a separate 144-item evaluation, the paper's PSCC method improved LLM-judged quality by 6.9 percentage points over A-Mem with roughly one-third fewer API tokens.

## Technical Skills

**Programming & data:** Python, R, SQL, Bash, Julia, PySpark, DuckDB, Polars, BQL
**Methods:** econometrics, factor models, causal inference, panel/time series, machine learning, NLP, computer vision
**Libraries & platforms:** NumPy, SciPy, scikit-learn, PyTorch, fixest, Databricks, Bloomberg, DevOps/Git, HPC/Slurm